

Image Processing

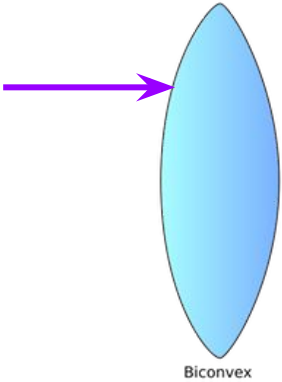
Image formation I:

thin lenses and optics tradeoffs

Francesco Solera

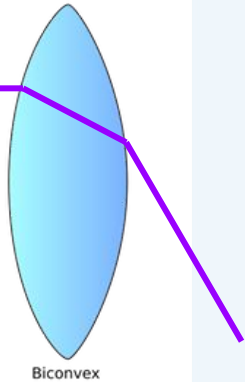
October 1st, 2026

Lenses and refraction

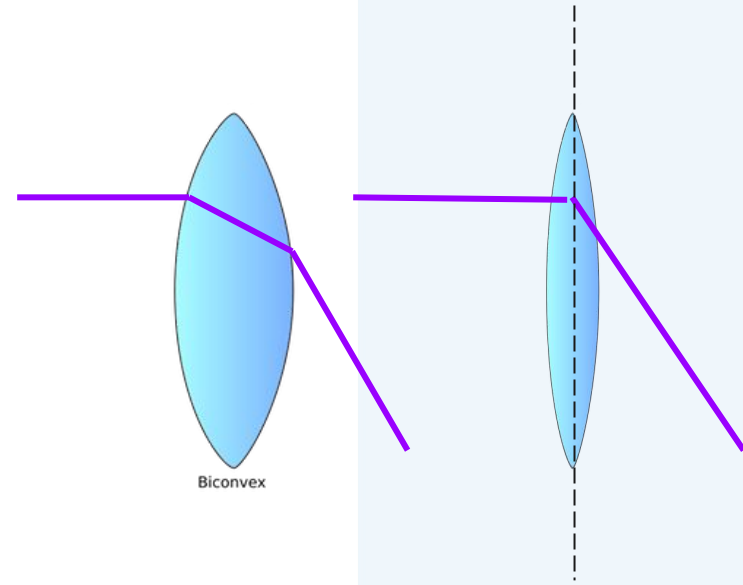


Lenses and refraction

Refraction occurs whenever light crosses a boundary between different refractive indices.



Lenses and refraction

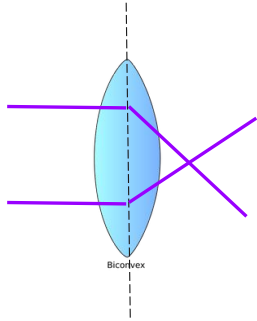


Refraction occurs whenever light crosses a boundary between different refractive indices.

In a standard setup this happens twice: when the ray of light enters the lens and when it exits the lens.

In a **thin lens** we assume that we can approximate the composition of bendings as a single one in the middle of the lens.

Convexity and densities

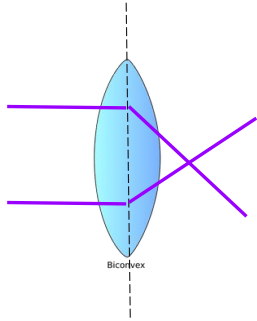


LOWER MEDIUM DENSITY

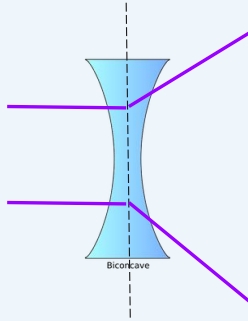
The most common case for a lens is to be **biconvex** and immersed in a less dense medium (i.e. air). In this case we will be observing a **convergent** behavior.

Convergence means different rays parallel to the optical axis will meet after traversing the lens.

Convexity and densities



LOWER MEDIUM DENSITY

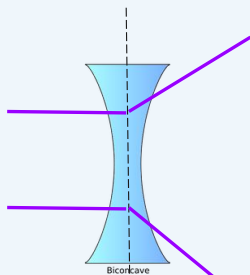
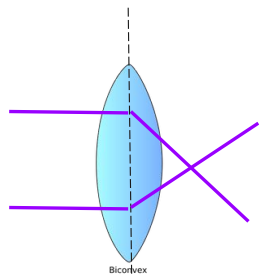


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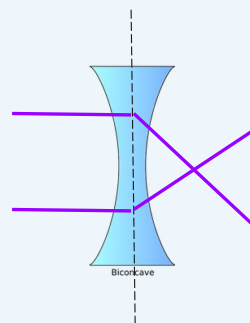
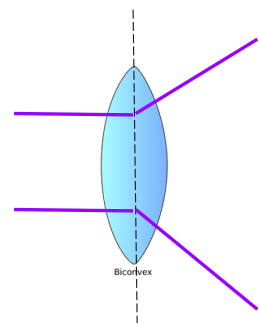
Convergence means different rays parallel to the optical axis will meet after traversing the lens.

For a **biconcave** lens in air we observe a divergent behavior instead.

Convexity and densities



LOWER MEDIUM DENSITY



HIGHER MEDIUM DENSITY

The most common case for a lens is to be **biconvex** and immersed in a less dense medium (i.e. air). In this case we will be observing a **convergent** behavior.

Convergence means different rays parallel to the optical axis will meet after traversing the lens.

For a **biconcave** lens in air we observe a divergent behavior instead.

But convergence also depends on the optical **density** of the lens and the external medium: if the external medium is optically denser than the lens, we observe the opposite behavior.

Snell's law

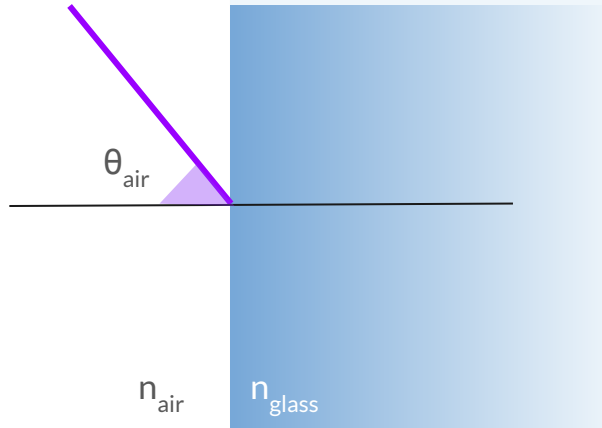
The way light bends is described by Snell's Law:

$$n_{\text{air}} \sin \theta_{\text{air}} = n_{\text{glass}} \sin \theta_{\text{glass}}$$

We consider the angles between the rays of light and the normal to the surface.

 n_{air} n_{glass}

Snell's law

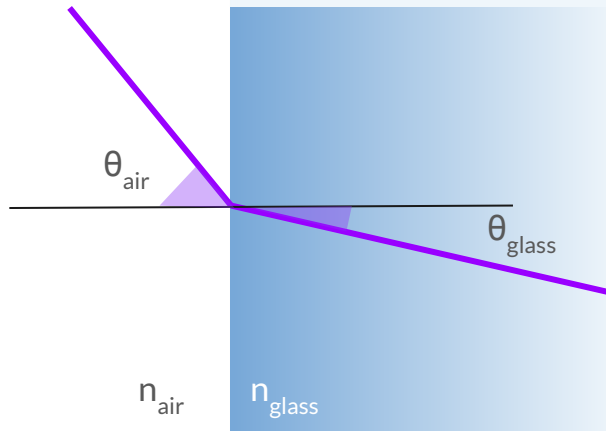


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Since sin function is increasing between 0 and $\pi/2$, for Snell's equation to remain balanced:

- if the ray enters a more optically dense material, the angle must decrease (as in the case with glass)
- and vice versa (e.g. if you trace a ray of light backwards)

Snell's law on spherical surfaces

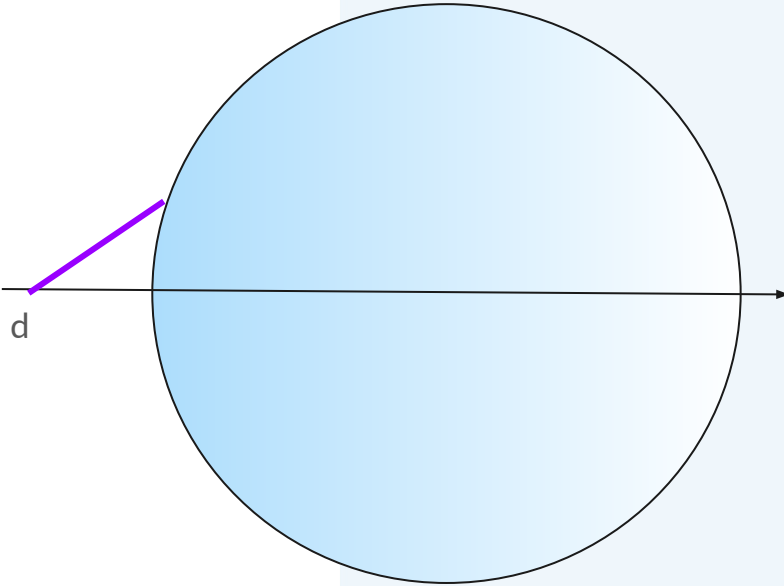
Under the following assumptions:



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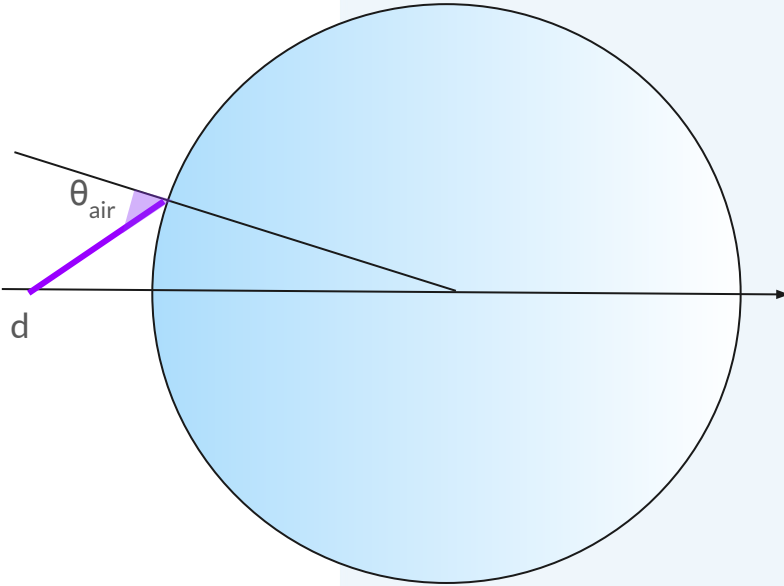
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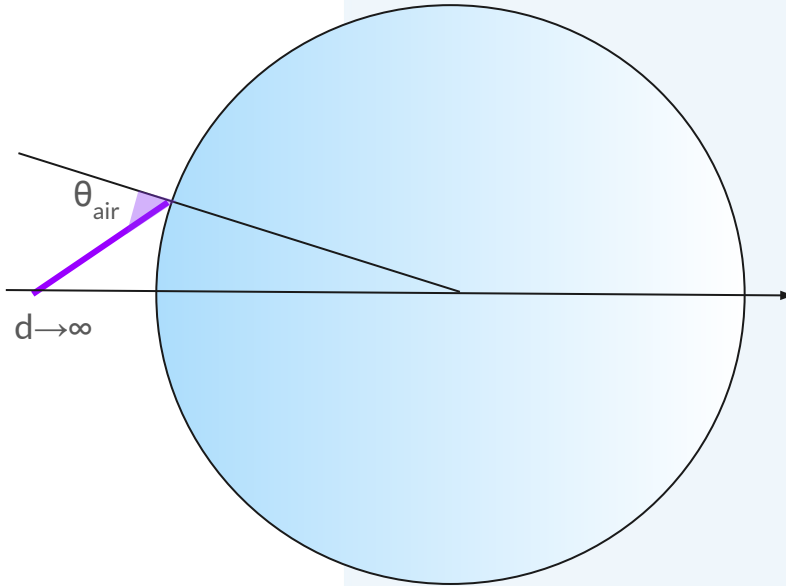
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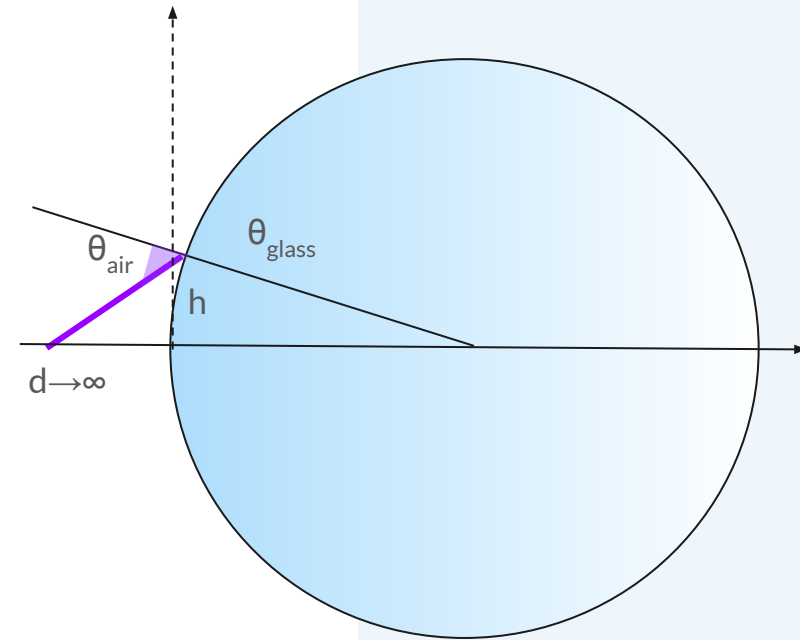
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Under the following assumptions:

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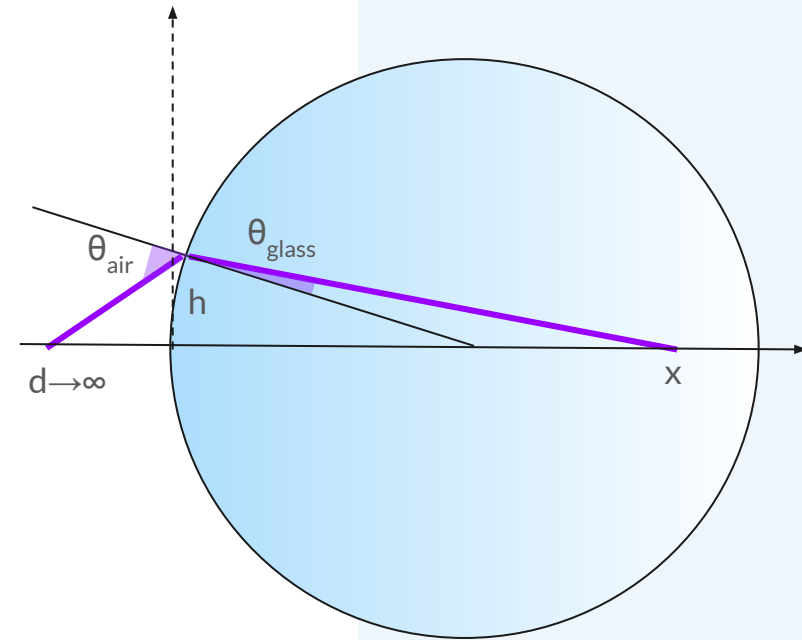
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Under the following assumptions:

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- object at infinite distance ($d \rightarrow \infty$, i.e. parallel ray)
- and assuming small angles involved (small h)

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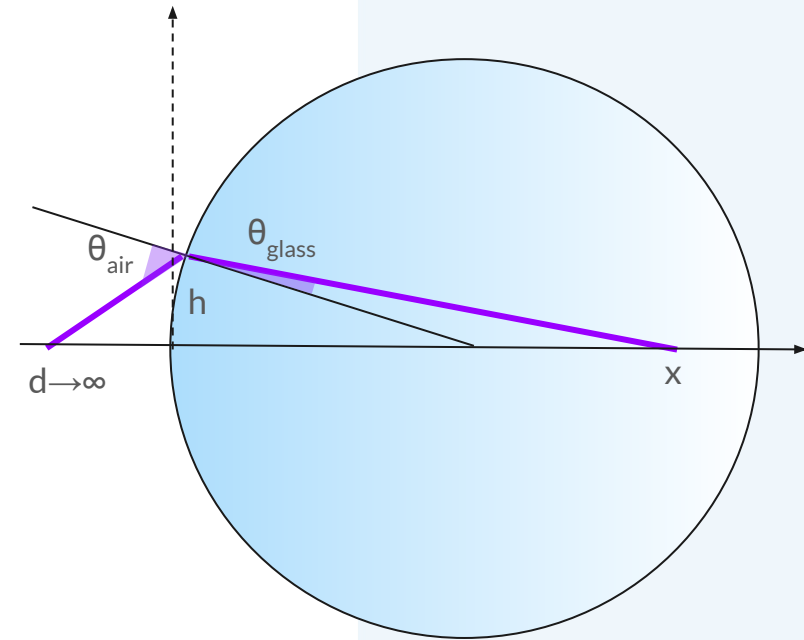
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we obtain:
$$x = \frac{n_2}{n_2 - n_1} R$$

(derivation is a bit long but not difficult if you're curious)

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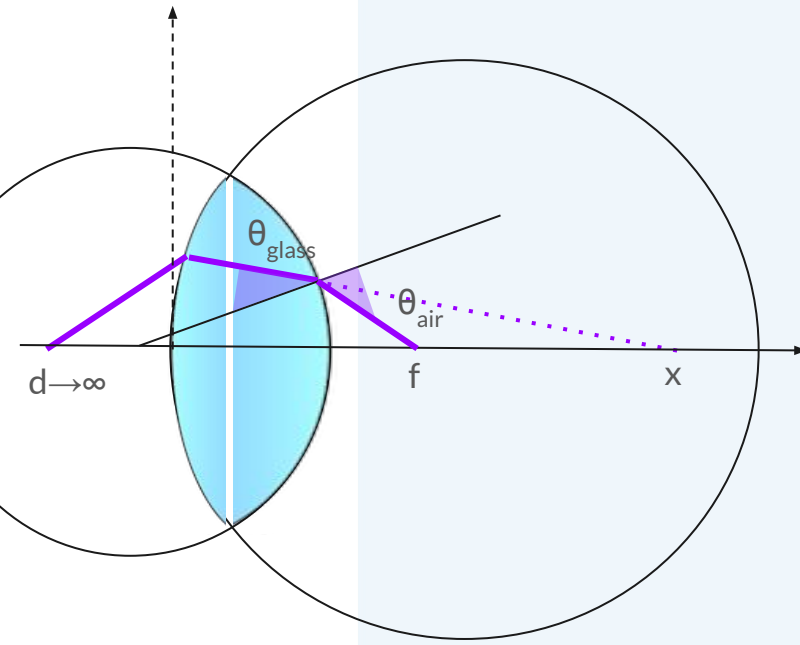
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It does not depend on h !

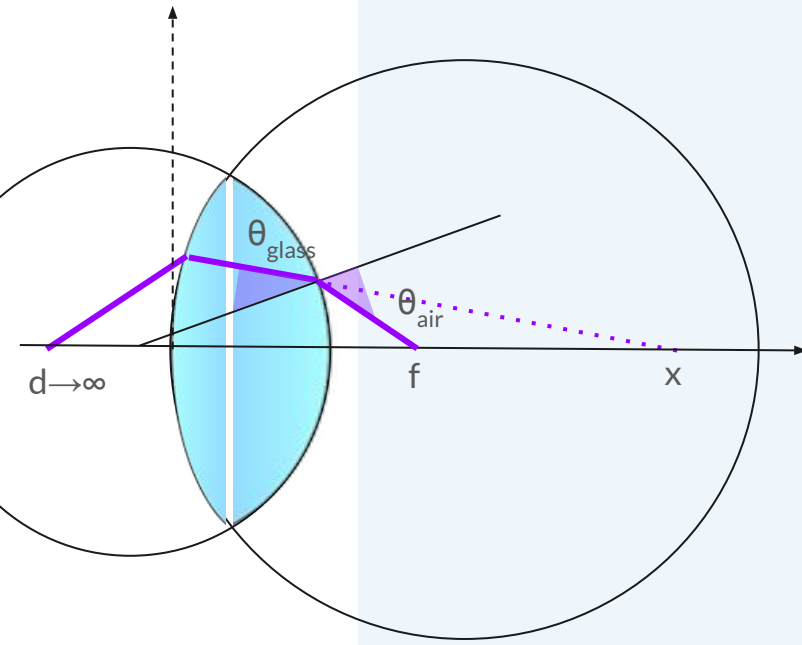
=> all parallel rays cross the optical axis at the same x

Lens maker equation



If we apply it twice, we obtain a formula that relates the shape of the lens with the point where all parallel rays converge on the optical axis, we call that point f .

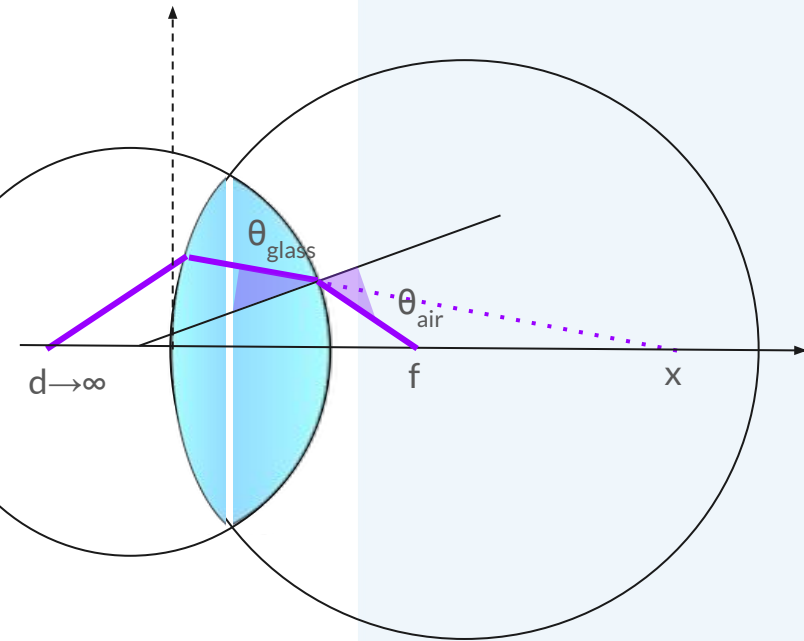
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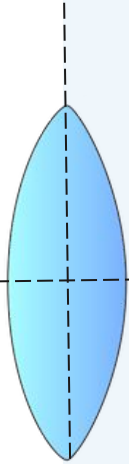
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By changing the curvature of the lens you can change f .

This formula is general, but we always assume standard biconvex lenses.

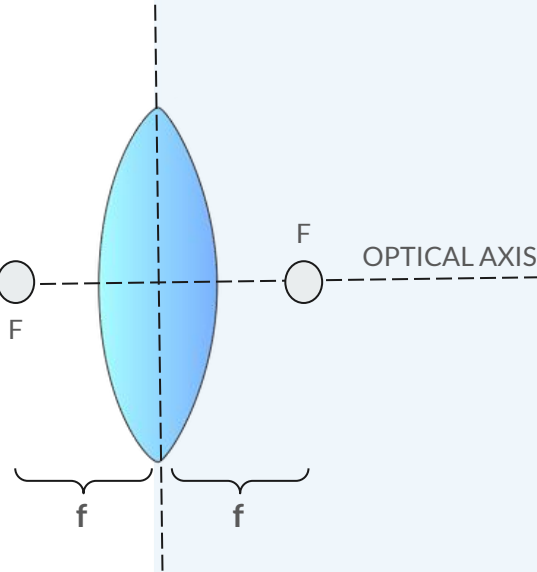
Focal length



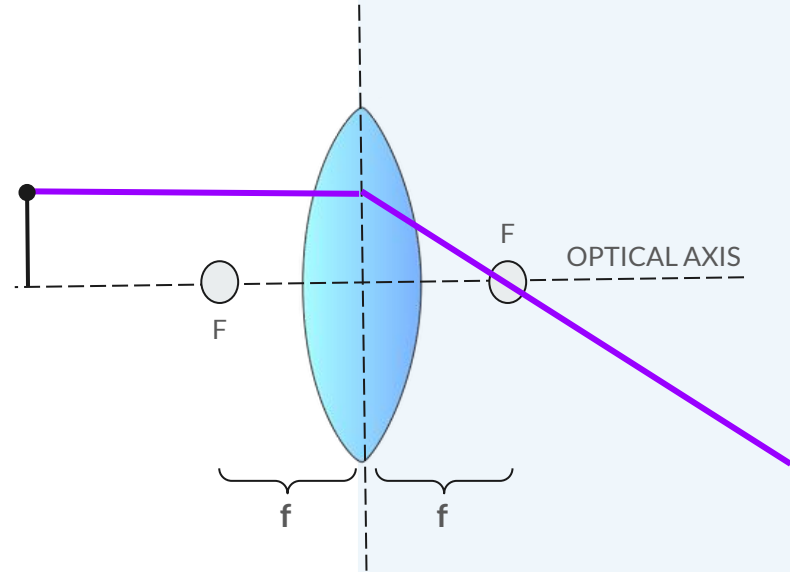
OPTICAL AXIS

Focal length

Each converging lens has a peculiar point called focal point F , where all rays of light parallel to the optical axis converge. It is found on both sides of the biconvex lens at a distance called **focal length f** .



Focal length

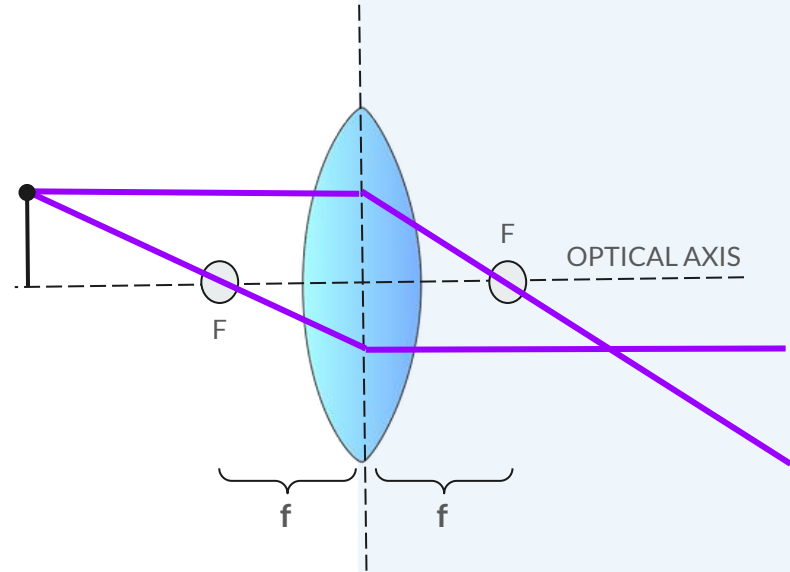


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When light bounces off an object in the scene it gets scattered in all directions, some of the rays go through the lens and we can neatly describe how 3 of them behave:

- the ray parallel to the optical axis

Focal length

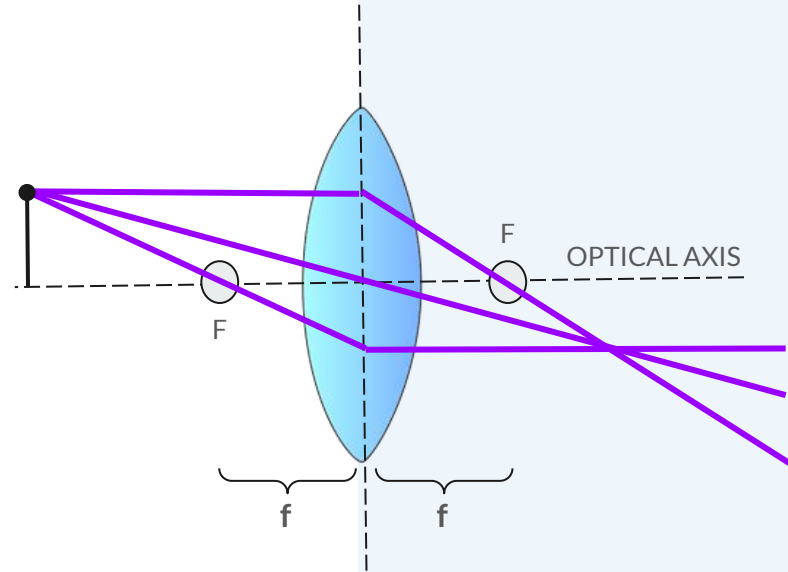


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Focal length



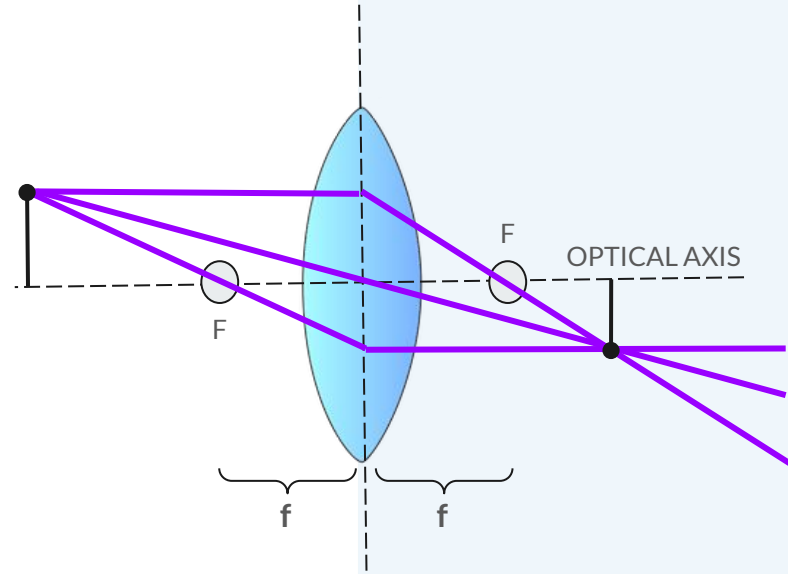
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- the ray parallel to the optical axis
- the ray passing through the focal point
- the ray passing through the center of the lens

(This last point follows from Snell's Law and the thin lens approximation: the two glass surfaces are parallel and very close, so the incoming and outgoing rays are very very similar...)

Focal length



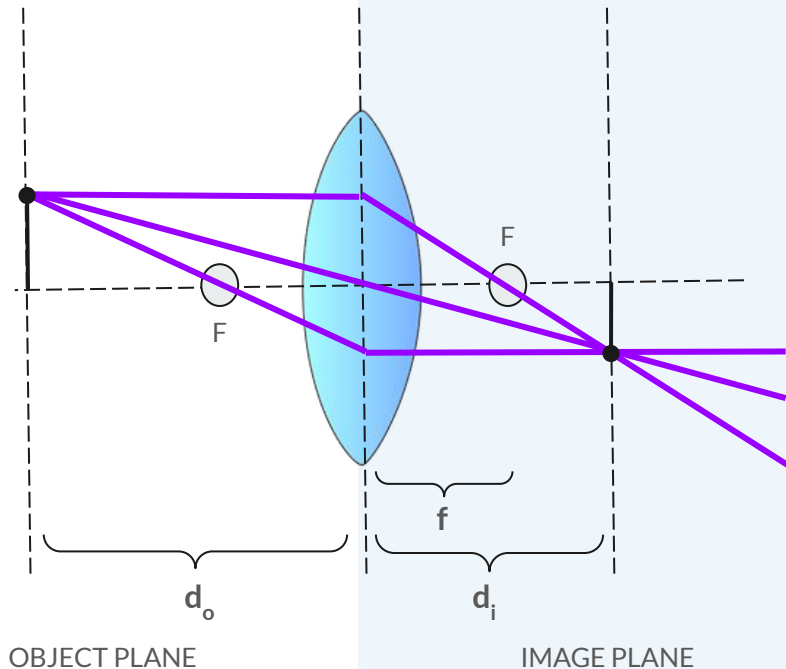
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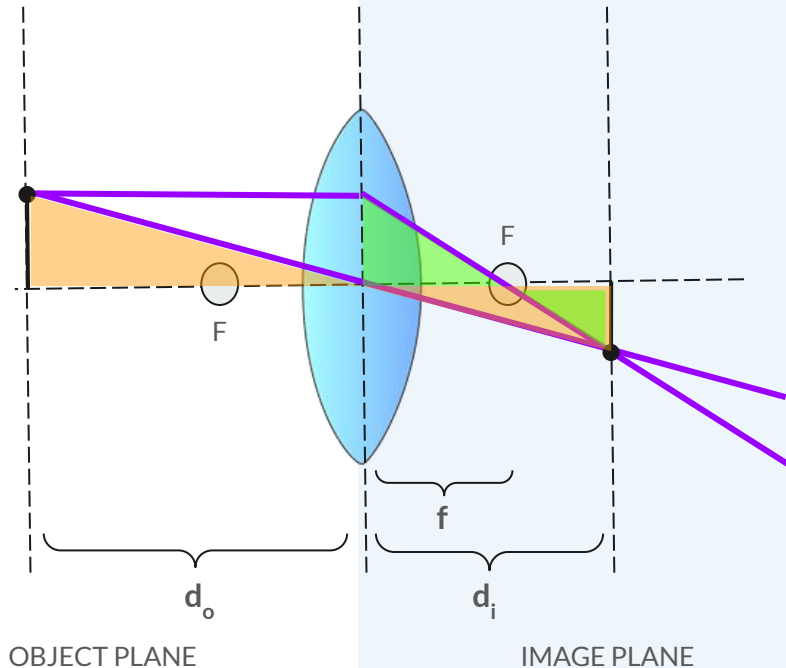
The focal length of a lens describes how objects to the left of the lens (real world) are transformed to the right of the lens (image world).

Thin lens equation



These 3 special rays will meet in the same spot at a distance d_i from the lens. The locus of points at this perpendicular distance from the lens is called **image plane**.

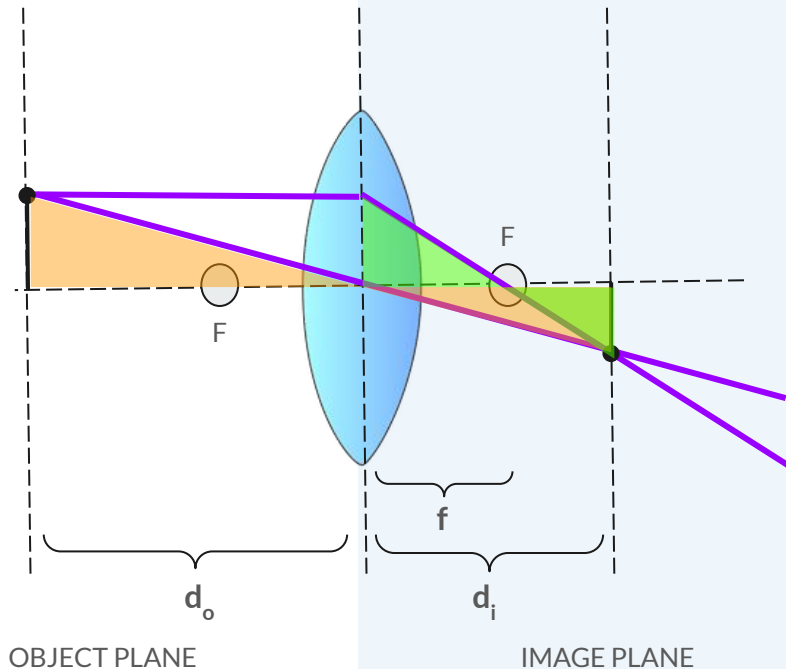
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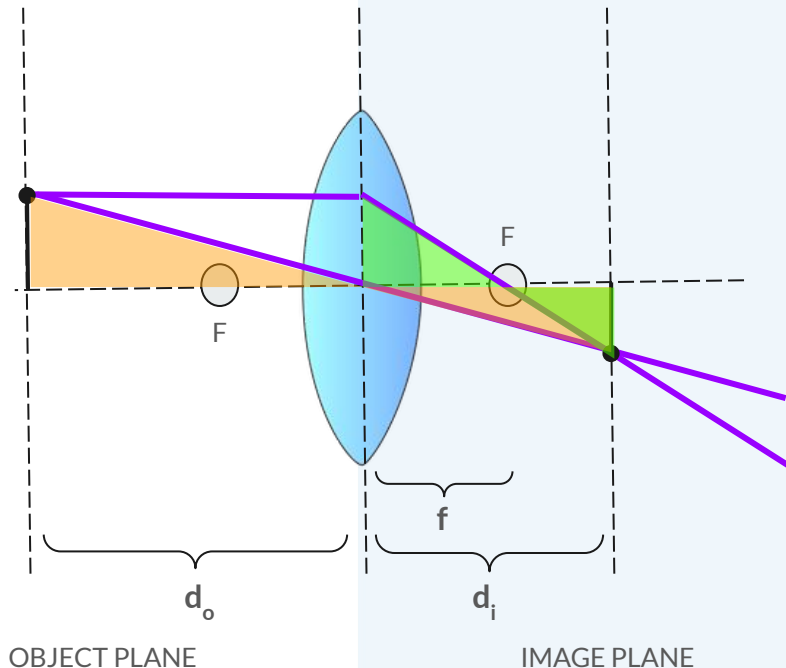
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$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

The **object plane** at distance d_o , the image plane at distance d_i from the lens and the **focal length f** are linked together by the **thin lens equation**.

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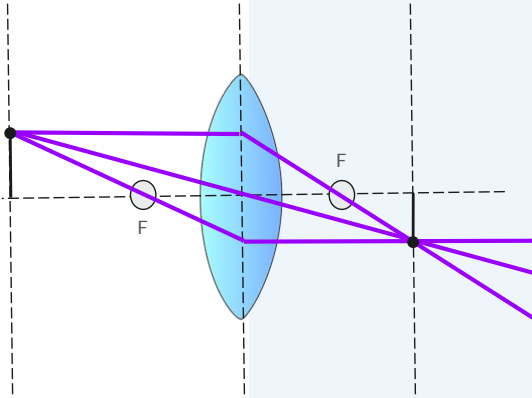
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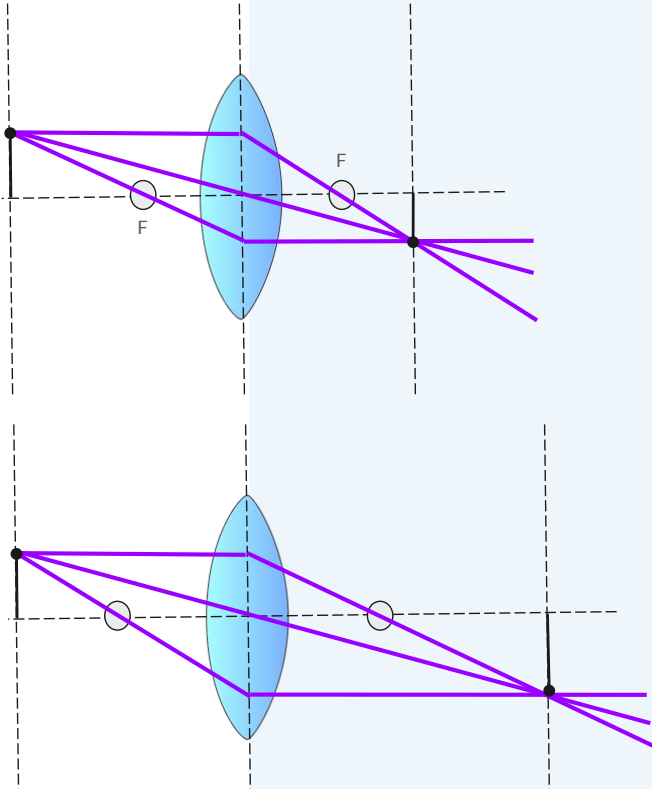
From this equation 2 effect follows: zoom and focus.

Magnification



The magnification is computed as $m = -d_i / d_o$

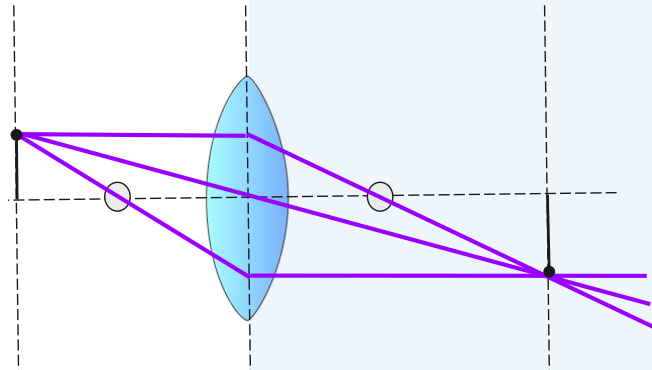
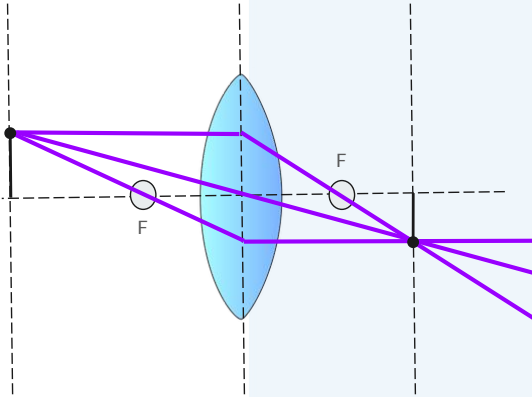
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If we keep d_o fixed and **increase f** , also d_i increases having a **zoom in** effect.

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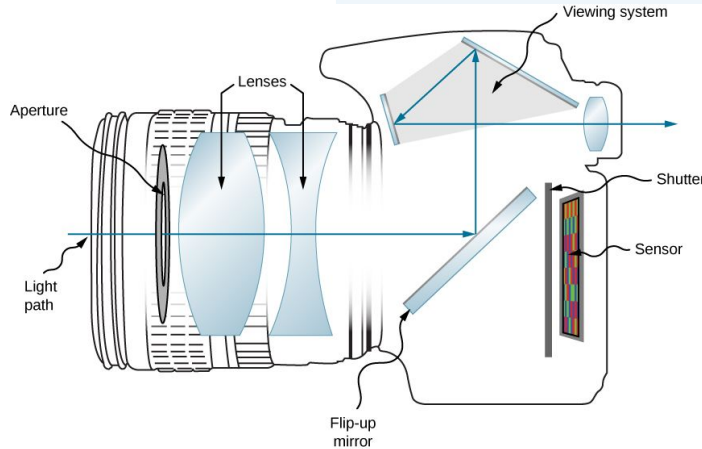
If we keep d_o fixed and **decrease f** , also d_i decreases having a **zoom out** effect.

Manual magnification



In consumer cameras, it is typically possible to change the zoom.

Manual magnification

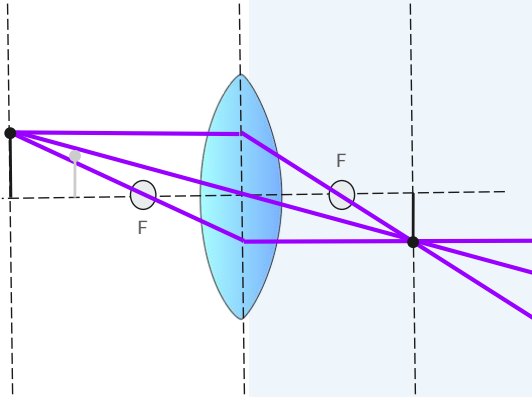


In consumer cameras, it is typically possible to change the zoom.

In fact, we are changing the focal length of the lens system by changing their relative configuration.

In industrial setup is very uncommon to have configurable focal length: during the design of the system you just select the most appropriate lens.

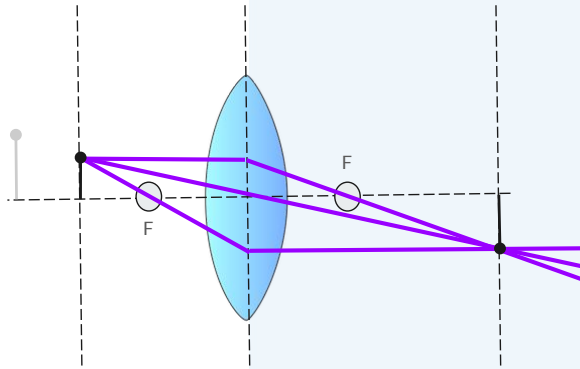
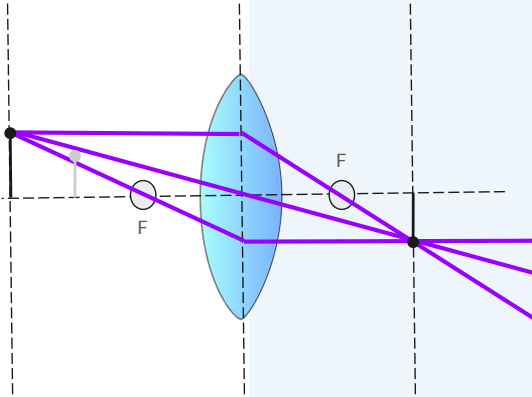
Focusing



The thin lens equation comes with a high price to pay: focus. The focal length constraints object distance and image plane distance.

With a standard biconvex lens, not everything can be in focus at once!

Focusing



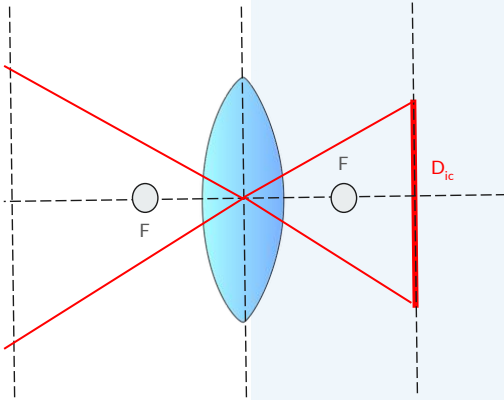
The thin lens equation comes with a high price to pay: focus. The focal length constraints object distance and image plane distance.

With a standard biconvex lens, not everything can be in focus at once!

If we **increase d_i** , d_o must decrease by the thin-lens equation, so we are **focusing closer** to the lens.

If we **decrease d_i** , d_o must increase by the thin-lens equation, so we are **focusing further** to the lens.

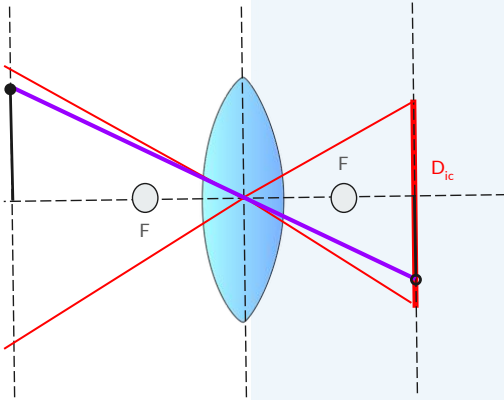
Maximum field angle



The focal length impacts the magnification of the lens but also the **maximum field angle**.

The maximum field angle is $\theta_{\max} = 2 \arctan(D_{ic} / 2d_i)$

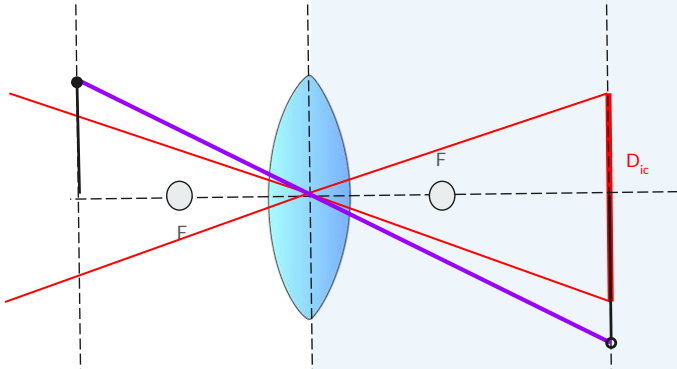
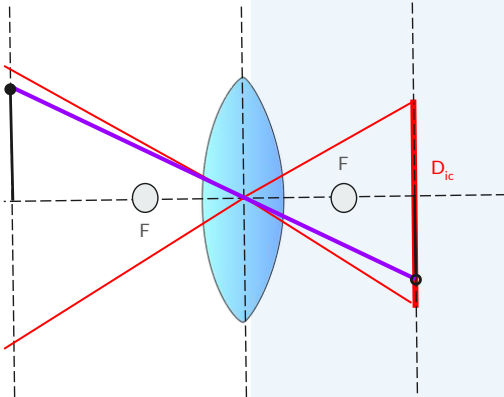
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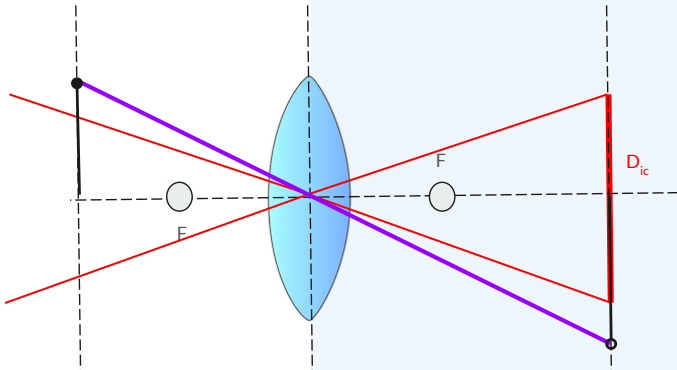
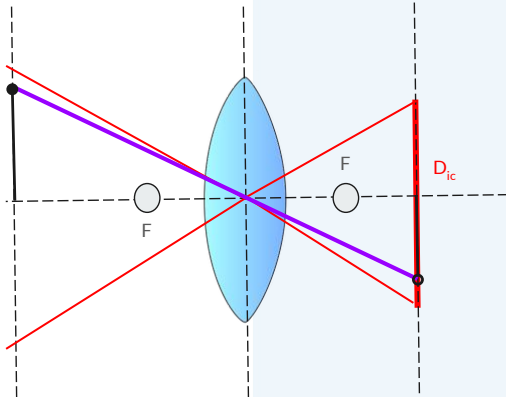


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The maximum field angle is $\theta_{\max} = 2 \arctan(D_{ic} / 2d_i)$

If we keep the image circle D_{ic} fixed, and **increase f** (thus increase d_i) we **restrict the maximum field angle**, and vice-versa.

Maximum field angle



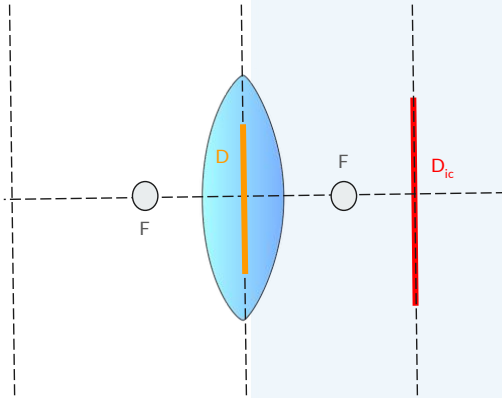
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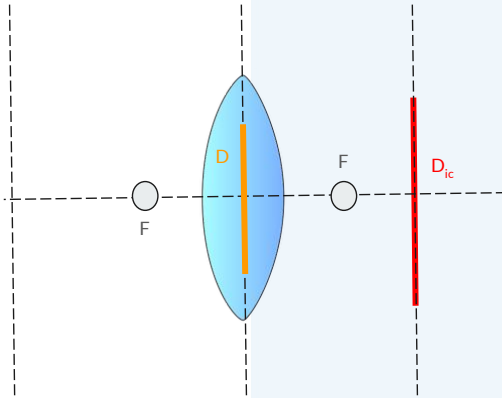
Magnification and maximum field angle always work together: a longer focal length zooms in but capture less scenes, a shorter focal length captures more of the scene with less zoom.

Aperture



Aperture (or *diaphragm*) limits the cone of rays that pass through the lens and reach the image plane.

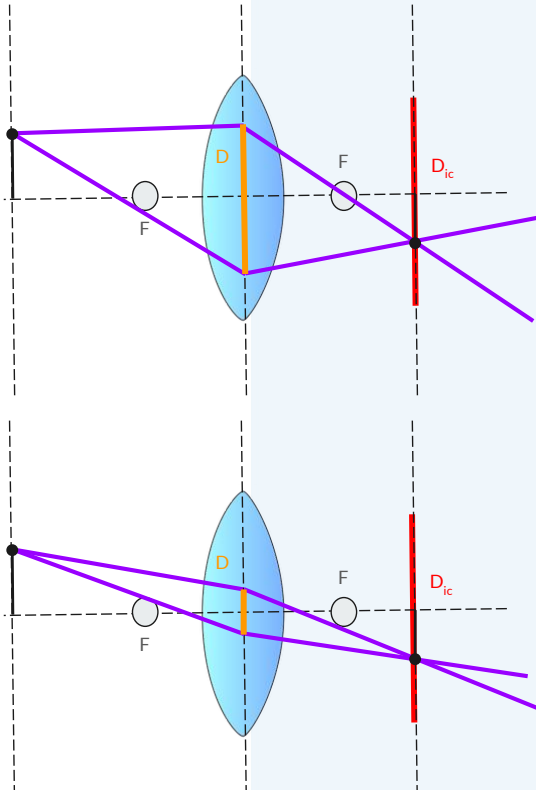
Aperture



Aperture (or *diaphragm*) limits the cone of rays that pass through the lens and reach the image plane.

First effect: we can use it to control the **brightness** of the image. Fewer rays of light means a darker image.

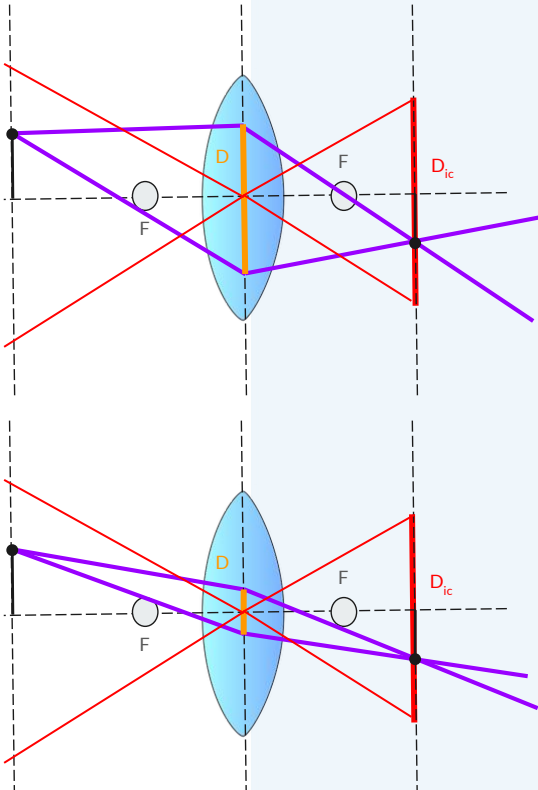
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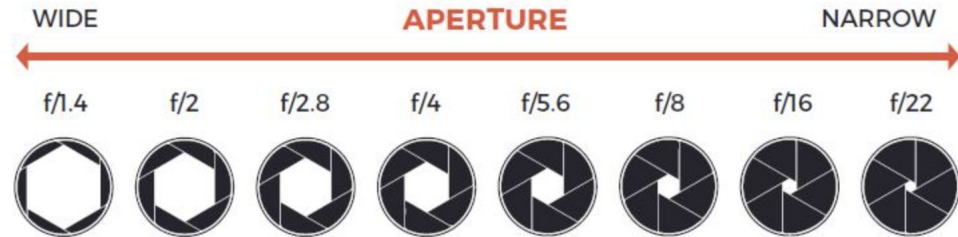
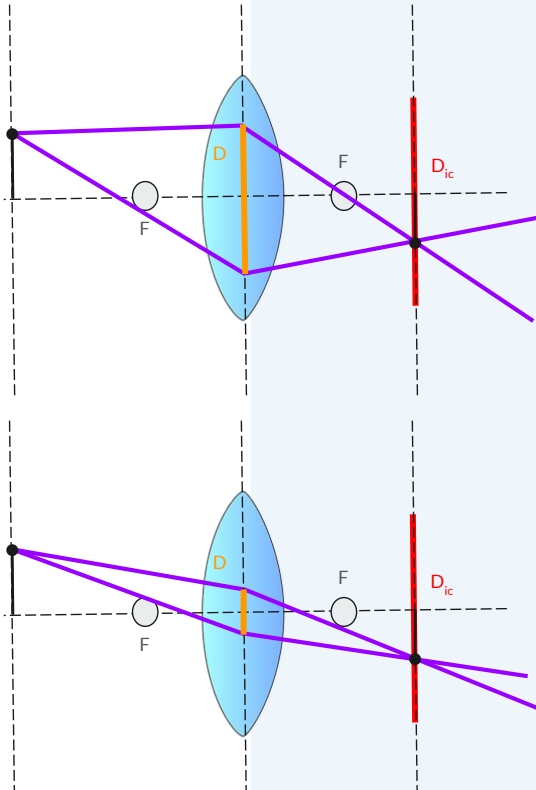


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Notably: in the ideal thin-lens model, the aperture does not influence the maximum field angle.

Aperture

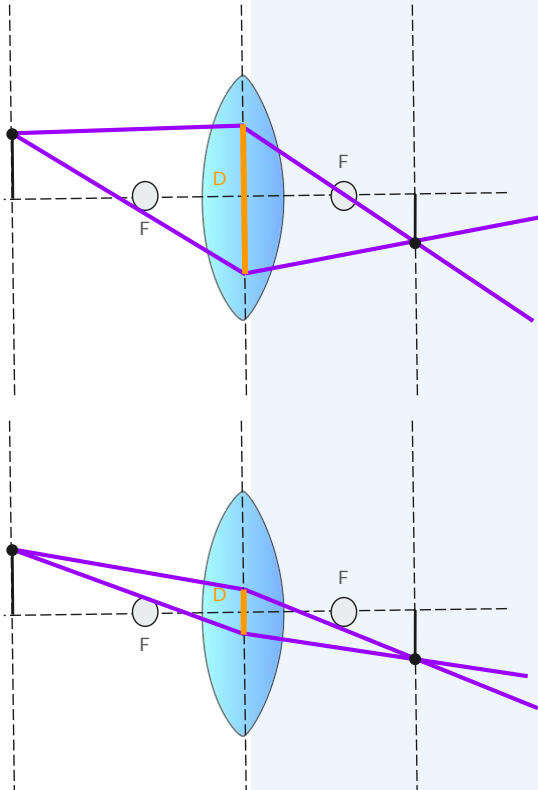


Aperture is often expressed through the **f-number**
 $N = f / D$

It is basically a ratio between the focal length and the pupil diameter.

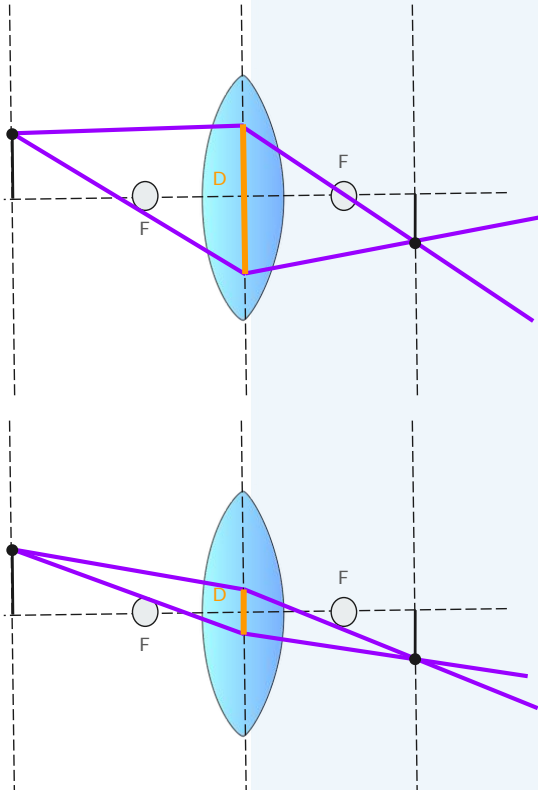
N are typically found in steps of $\sqrt{2}$, called **f-stops**. Increasing the f-number by a factor of $\sqrt{2}$ halves the irradiance; decreasing it by $\sqrt{2}$ doubles it.

Depth of focus



Second effect of smaller aperture: rays diverge slower away from the image plane

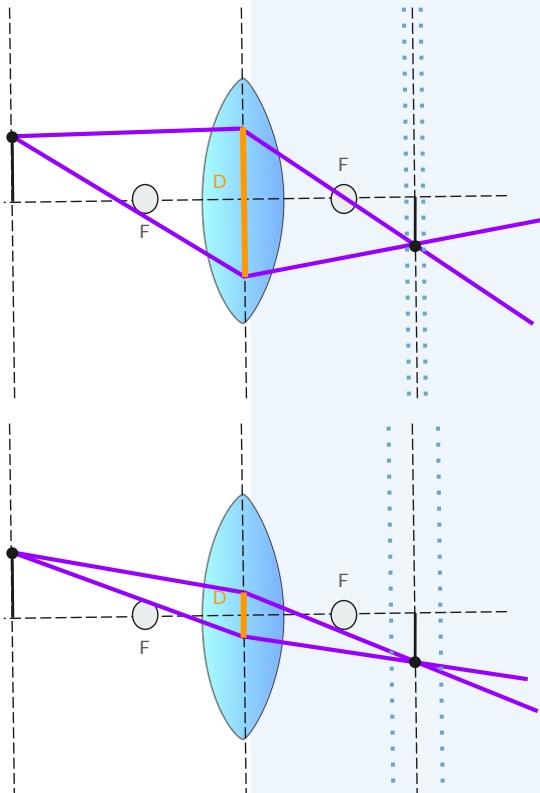
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Being out of focus, means that lights does not converge to a point but to a disk, i.e. the **circle of confusion** (coc).

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Second effect of smaller aperture: rays diverge slower away from the image plane, i.e. the image focus degrades slower away from d_i' .

Being out of focus, means that lights does not converge to a point but to a disk, i.e. the **circle of confusion** (coc).

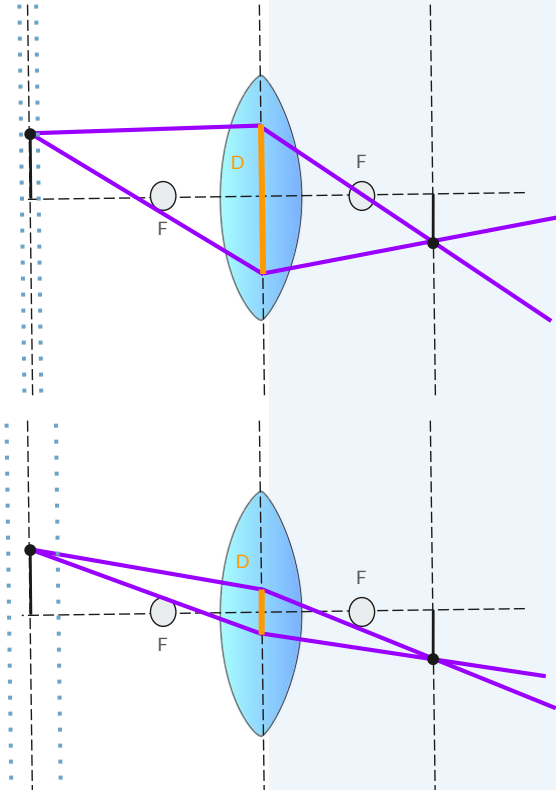
Depth of focus is the image-space interval over which the blur diameter remains below a chosen circle-of-confusion threshold.

The **depth of focus increases** also when we are focusing things **closer to the lens**, i.e. for larger d_i the cone of light rays shrinks.

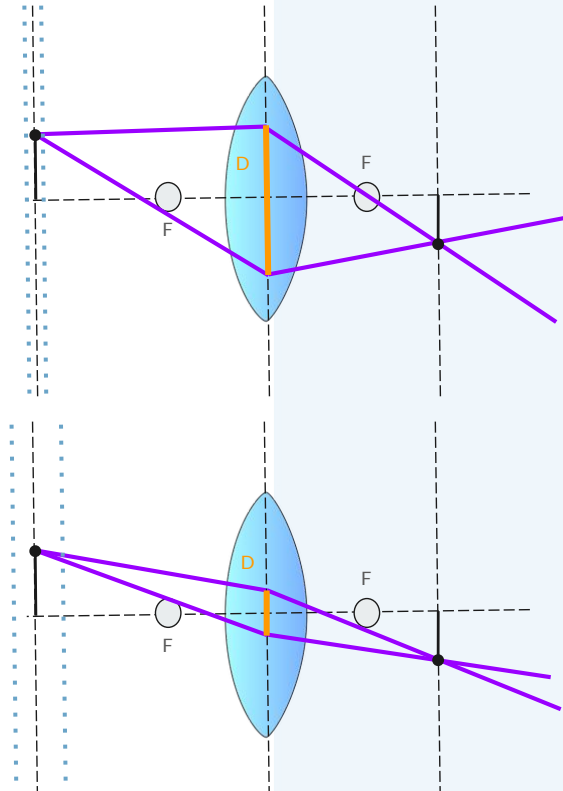
Depth of field

Light might not converge to a point because we are measuring away from d_i or, analogously, we are observing things that are not at a distance d_o .

Depth of field: range around d_o in object space where lights converges to a coc that is considered acceptable.



Depth of field



Light might not converge to a point because we are measuring away from d_i or, analogously, we are observing things that are not at a distance d_o .

Depth of field: range around d_o in object space where lights converges to a coc that is considered acceptable.

The **depth of field increases** also when we are focusing things **further from the lens**, i.e. for larger d_o the cone of light rays shrinks.

Depth of focus and depth of field are two faces of the same coin, deeply linked by the thin-lens equation.

Diffraction and Airy disk

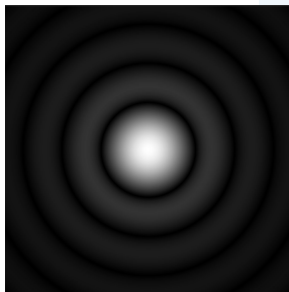
Ideally, we could make the aperture as small as possible to have almost infinite depth of field.

Diffraction and Airy disk

Ideally, we could make the aperture as small as possible to have almost infinite depth of field. In practice, this doesn't yield a clean image due to a competing factor: diffraction.

Diffraction is the spreading of a wave, light in our case, after passing through a finite aperture.

Diffraction and Airy disk



Ideally, we could make the aperture as small as possible to have almost infinite depth of field. In practice, this doesn't yield a clean image due to a competing factor: diffraction.

Diffraction is the spreading of a wave, light in our case, after passing through a finite aperture.

Interferences of different light rays create what is called **Airy disk** of radius r and denotes the physical clarity limit of the lens system. Diffraction is a fundamental limit even for a perfect lens. It's always present, but reducing the aperture increases the interferences.

Effects of diffraction and depth of focus/field combine together.

$$d_{\text{Airy}} = 2.44\lambda N$$

Interactive visualization



IMAGE PROCESSING
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IMAGE FORMATION
Thin lenses

L1 Thin lenses
 L2 Aperture and focus
 L3 Spatial sampling
 L4 Temporal sampling

STEP-BY-STEP TUTORIAL
Balance the equation
 Can you predict the third term? Keep f fixed and bring the object closer; watch d_i grow and the image enlarge. Then change what is fixed to test how any two terms determine the third.

< Previous Step 3 of 8 Next >

Parameters

Focal length f ☒ Fix
 distance between focal point F and optical center of the lens
 40 mm

Object distance d_o ☐ Fix
 distance between the object and the optical center of the lens
 120 mm

Image distance d_i ☐ Fix
 distance between the image plane and the optical center of the lens
 60 mm

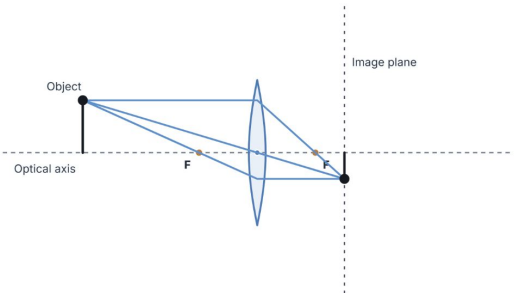
Object height h_o
 perpendicular distance of the object from the

THIN-LENS EQUATION

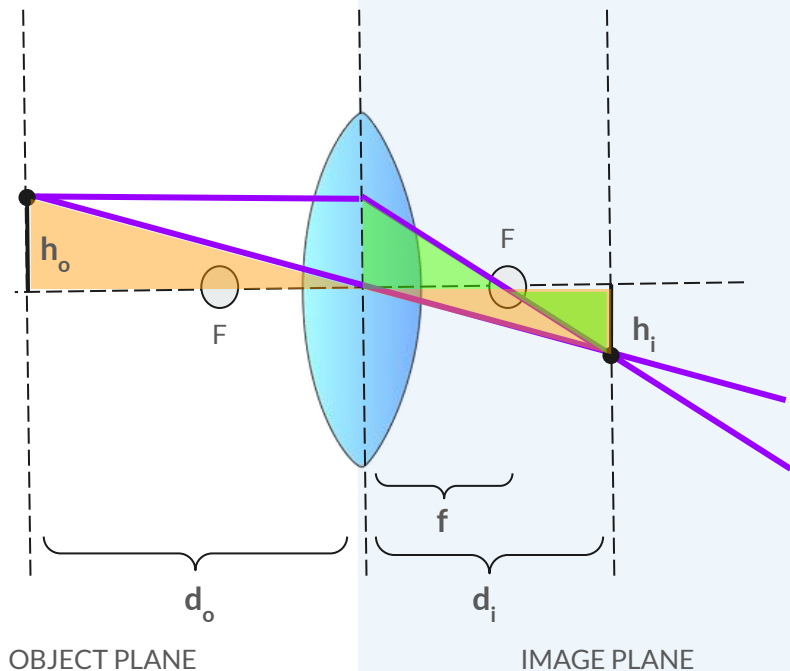
$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

IMAGE PLANE FRONT VIEW


Trace light rays Parallel Central Focal



Appendix A: thin lens equation



We can define two set of similar triangles.

Central ray triangles (orange):

$$h_i/h_o = d_i/d_o$$

Focal ray triangles (green):

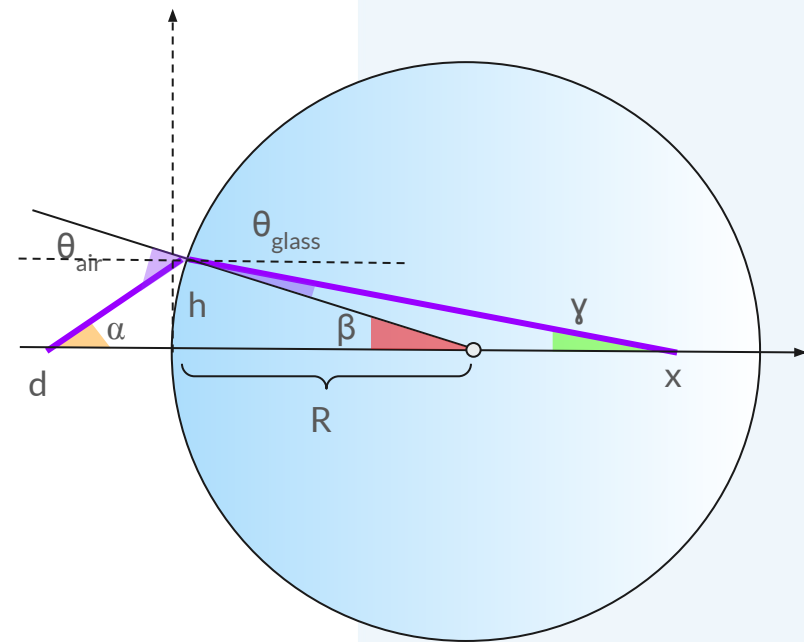
$$h_i/h_o = (d_i-f)/f$$

$$\text{Thus: } d_i/d_o = (d_i-f)/f$$

By rearranging, we obtain the thin-lens eq:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

Appendix B: refraction at a spherical surface



Let $\bar{\alpha}$, $\bar{\beta}$ and $\bar{\gamma}$ be the $\pi/2$ complement to α , β and γ respectively.

We can compose θ_{air} from the sum of α and β (both by similar triangles). And we can obtain θ_{glass} by subtracting $\bar{\beta}$ from $\bar{\gamma}$.

If h is small, θ_{air} and θ_{glass} are also small, thus $\sin(\theta) \approx \tan(\theta) \approx \theta$. Thus we can approximate these angles through their side lengths:

$$\theta_{\text{air}} = \beta + \alpha = h/R + h/d$$

$$\theta_{\text{glass}} = \bar{\gamma} - \bar{\beta} = \pi/2 - \gamma - \pi/2 + \beta = h/R - h/x$$

Under the additional approx. that $d \rightarrow \infty$ (parallel ray): $\theta_{\text{air}} = h/R$

If we now substitute into the approx. Snell's law $n_1 \theta_{\text{air}} = n_2 \theta_{\text{glass}}$, we obtain: $n_1(h/R) = n_2(h/R - h/x)$. If we re-arrange, we obtain a formula where the h have canceled out (key for the definition of focal length):

$$x = \frac{n_2}{n_2 - n_1} R$$